ABSTRACT: This study investigates the role of the intact superior laryngeal nerve (SLN) in the reinnervation process of one of the laryngeal muscles, the posterior cricoarytenoid muscle (PCA), following recurrent laryngeal nerve (RLN) injury. Using a chronic RLN injury model in the adult rat, PCA reinnervation was assessed by retrograde double-tracing techniques in combination with electrophysiology and immunohistochemistry of muscle sections. The results demonstrate that the PCA receives dual innervation from both laryngeal nerves even in the uninjured system. Functionally significant collateral reinnervation originates from intact SLN fibers following RLN injury, mainly due to intramuscular sprouting rather than by recruitment of more motor neurons. This may be important when choosing surgical and/or medical treatment for patients with RLN injury.
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COLLATERAL REINNERVATION BY THE SUPERIOR LARYNGEAL NERVE AFTER RECURRENT LARYNGEAL NERVE INJURY

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The larynx is innervated by two branches of the vagus nerve: the recurrent laryngeal nerve (RLN) and the superior laryngeal nerve (SLN).[3,20] The RLN has traditionally been regarded as the main motor nerve for the intrinsic laryngeal muscles, whereas the SLN has been thought to provide motor fibers exclusively to the cricothyroid muscle (CT).[40] This classical view of human laryngeal neuroanatomy has, however, been questioned in several anatomical dissection studies. Those studies have shown that branches from the SLN seem to innervate both the thyroarytenoid (TA) and the posterior cricoarytenoid (PCA) muscles.[23,24,33,41] It is also known that injection of horseradish peroxidase into the internal branch of the SLN in resected human larynges results in anterogradely labeled axons not only in the CT, but also in the TA and PCA muscles,[42] bringing the classical neuroanatomical picture further into question. The reportedly small motor unit sizes of the laryngeal muscles[10,32] reflect the requirement for precise innervation.

Clinical injury to the RLN may lead to various degrees of vocal cord motion impairment, secondary to ineffective, unsynchronized (synkinetic) muscle activity rather than chronic denervation and marked muscle atrophy.[8] This indicates that pathological reinnervation of the intrinsic laryngeal muscles has occurred, and it is not in line with normal laryngeal physiology. The reason for this non-functional laryngeal reinnervation following RLN injury may be the loss of adductor- and abductor-fiber organization during regenerative growth of the RLN[13] and the subsequent misguided reinnervation of the muscle groups. If this is indeed the case, it would be analogous to the situation after facial nerve transection injury, where misdirected regeneration into the different nerve branches is considered to be a major pathological factor.[1,15] There has also been some speculation that intact nerves in the vicinity of the RLN (i.e., superior laryngeal nerve, contralateral RLN, pharyngeal nerve, cervical plexus) may innervate laryngeal muscles that have been deprived of RLN innervation.[8]

Following partial denervation of a motor nerve, the remaining axons have the ability to expand their territories and reinnervate the muscle by collateral axonal sprouting.[5–7,37] The sprouts emerge at the nodes of Ranvier or from the nerve terminals, a

Abbreviations: AChR, acetylcholine receptor; CMAP, compound muscle action potential; CT, cricothyroid muscle; CTB, cholera toxin subunit B; EMG, electromyogram; FG, fluorogold; NA, nucleus ambiguus; NF, neurofilament; PCA, posterior cricoarytenoid muscle; RLN, recurrent laryngeal nerve; SLN, superior laryngeal nerve; SP, synaptophysin; TA, thyroarytenoid muscle

Key words: electrophysiology; laryngeal synkinesis; nucleus ambiguous; posterior cricoarytenoid muscle; retrograde tracing

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process that depends on trophic and mechanical dynamics between the axon, the denervated muscle, and the perisynaptic Schwann cell.[6,9,38,39] The motor units are thus enlarged, and become able to compensate for up to 80–85% of original axonal loss.[14,22,30] In this case, the function of the target organ may be nearly normal. However, if collateral reinnervation is provided by another, adjacent peripheral nerve, the result would be a more pronounced alteration of the neuroanatomy. The SLN is connected to the RLN by several anastomoses.[34] Following RLN injury, this relation may serve as a base for collateral reinnervation by intact SLN fibers. Sprouting from intact SLN fibers could theoretically occur within the epineurial sheath in the anastomosis network or from peripheral, intramuscular sprouting within the denervated muscle compartment.

The aim of the present study was to investigate whether the SLN contributes to reinnervation of the denervated PCA. The PCA is the sole abductor of the vocal cords, and it is regarded classically as being innervated only by the RLN.[3,20,40] The innervation was assessed by: (1) retrograde double-labeling and quantification of the brainstem neurons that project through the SLN to the PCA muscle; (2) electrophysiological evaluation of the SLN and RLN innervation of the PCA muscle; and (3) visualization and qualitative analysis of the neuromuscular junctions in the PCA muscle.

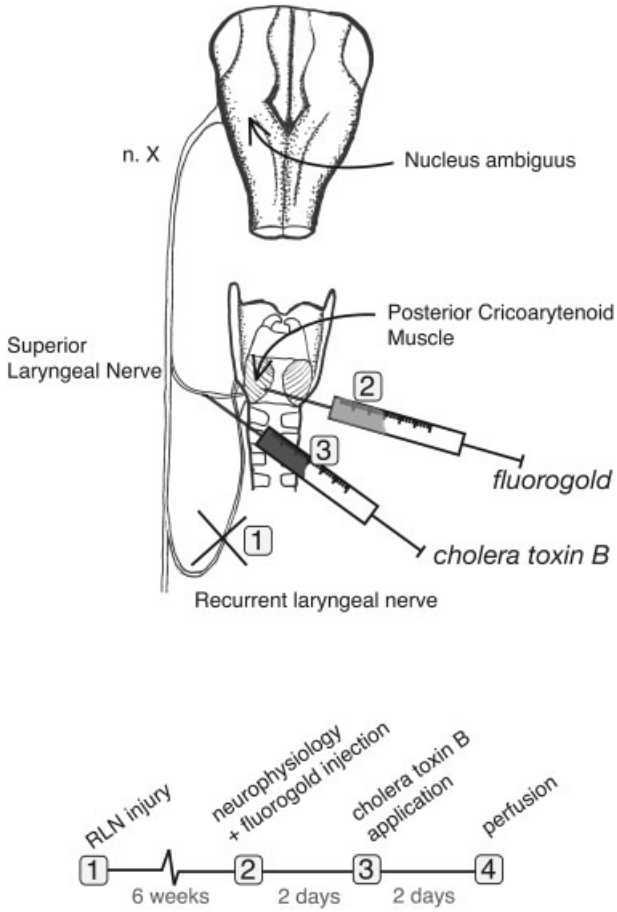


FIGURE 1. Schematic drawing of the brainstem and larynx (posterior view), demonstrating the surgical methods.

METHODS

Experiments were carried out with the approval of the Swedish Animal Welfare Agency. Group sizes were decided based on previous experience that showed neurobiologically important differences are detectable in groups of at least 6 animals. A total of 18 adult Sprague-Dawley rats (250–270 g) were used. Twelve of the animals were subjected to chronic RLN lesion. At 6 weeks post-lesion, electrophysiological evaluation was performed to evaluate the innervation of the PCA muscle ($n = 12$). Retrograde labeling of neurons projecting to the PCA muscle through the SLN was then performed in 6 of these animals ($n = 6$; Fig. 1), whereas the remaining 6 animals were further processed for muscle morphological analyses ($n = 6$). In a control group ($n = 6$), electrophysiology and retrograde tracing were performed without the preceding RLN injury. The groups were compared using unpaired *t*-tests.

Surgery. The rats were anesthetized by intraperitoneal injection of ketamine (50 mg/kg body weight; Ketalar; Pfizer) and medetomidine (0.5 mg/kg body weight; Domitor; Orion Animal Health). The animals underwent surgery at three different timepoints, as described in what follows.

RLN Lesion. The left RLN was identified and cut at the level of the seventh tracheal ring (Fig. 1). To obtain a chronic nerve lesion and complete unilateral deprivation of RLN input to the larynx, the proximal end was ligated with a 10/0 suture and directed away from the larynx, toward the thoracic aperture.

Electrophysiological Evaluation and Tracing of PCA Neurons. The rats were re-anesthetized, and the left SLN was identified. The sternothyroid muscle on the left side was used to gently lift the larynx and expose the PCA muscle. A single 0.51-mm steel hook-wire electrode (Viasys Healthcare, Madison, Wisconsin) was inserted into the body of the left PCA in a rostral-to-caudal direction. The reference electrode was placed at the rostral end of the wound and connected to a Keypoint workstation (Medtronic,

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Tolochenaz, Switzerland). An electromyogram (EMG) was recorded; in normal controls, this showed a phasic, inspiratory pattern synchronous to PCA contraction. Subsequently, the left SLN was stimulated at a point 5 mm from its entry into the larynx, using a 1-mm needle electrode, at supramaximal intensity (0.5 mA, stimulation duration 0.1 ms). In addition, the left RLN was stimulated at the level of the eighth tracheal ring. The compound muscle action potential (CMAP) was recorded from the left PCA. Following electrophysiological examination, 0.2 μl of 5% (w/v) fluorogold (FG; Fluorochrome, Inc., Denver, Colorado) was pressure-injected into the left PCA through a glass micropipette without visual leakage, and the area was rinsed with saline.

Tracing of SLN Neurons. Two days after the FG injection, the rats were re-anesthetized, and the larynx was again exposed. The left SLN was identified and cut at a location just before it entered the larynx. The proximal end was inserted into a polyethylene capsule containing a solution of 1% cholera toxin subunit B low salt (CTB; List Biological Laboratories, Campbell, California) and left there for 15 minutes. The capsule was then removed, and the area was rinsed with saline. Two days later, the animals were deeply anesthetized and euthanized by intracardiac perfusion of 200 ml of body-temperature saline, followed by 250 ml of cold paraformaldehyde (4%) in phosphate buffer.

Tissue Processing. The brainstem and the larynx were removed and postfixed in the same fixative for 1.5 hours, then rinsed in saline and transferred to phosphate buffer (pH 7.4) containing 15% (w/v) sucrose overnight. Brainstems were cut transversely in series of 20 + 14 + 14 μm using a Leica CM3000 cryostat at −24°C. A total of 90 20-μm sections were collected, covering a rostrocaudal distance of 90 × (20 + 28) μm = 4.32 mm, enveloping the area of interest in the nucleus ambiguus (NA). The larynx was symmetrically mounted, and the PCA muscle was cut in 14-μm frontal sections.

Immunohistochemistry. Brainstem and muscle sections were initially incubated with bovine serum albumin 1%, 0.1% Na-Azid, and 0.3% Triton X-100 (Sigma Co., St. Louis, Missouri) in phosphate buffer for 60 minutes at room temperature.

To visualize CTB, brainstem sections were then incubated overnight at 4°C in phosphate buffer containing 0.3% Triton X-100 with the antibody goat anti-choleragenoid 1:10,000 (List Biological Laboratories, Campbell, California). To visualize immunoreactivity, sections were processed with Cy3-conjugated secondary antibody for 1 hour at room temperature. The sections were mounted in a 1:1 solution of glycerol and phosphate buffer.

Muscle sections were incubated overnight at 4°C in phosphate buffer containing 0.3% Triton X-100 with the following primary antibodies: (1) monoclonal anti-human neurofilament protein 1:100 (Dako-Cytomation, Denmark) and (2) anti-synaptophysin g9.5 1:10 000 (a gift from Dr. R. Jahn, Max Planck Institute, Leipzig, Germany). Sections were then incubated for 1 hour at room temperature with secondary antibodies conjugated to Cy3 and Cy5, respectively (Jackson Labs). Subsequently, muscle sections were processed for 2 hours at room temperature with Alexa-488–conjugated α-bungarotoxin (Molecular Probes, Eugene, Oregon) to visualize acetylcholine receptors.

Representative PCA muscle sections were also chosen for routine hematoxylin (Meyer) and eosin staining, and mounted in a non-aqueous matter.

Quantification of Retrogradely Traced PCA Neurons. Fluorogold and Cy3 immunoreactivity for CTB were visualized using a Leica DM 4000B fluorescence microscope (lenses: 20× and 40×). The two tracers were seen as brightly assembled in the cytoplasm but respecting the cell nucleus (Fig. 2). Positively stained cells were counted manually in all serially cut 20-μm sections; that is, 42% of the nucleus ambiguus. Because of split cell bodies at the section surfaces, only those neurons with a clearly visible nucleus were counted, and thus it was not possible to calculate the total number of neurons in the NA; this estimation would have required a stereological approach. Shrinkage of the brainstem due to neuronal loss may result in counting errors in serial sections. This phenomenon was not adjusted for, because our earlier results showed that a similar injury (RLN resection) in the same system causes no neuronal loss.[18]

Analysis of PCA Muscle Sections. Muscle sections were analyzed using a Leica confocal system (TCS SP2) connected to a Leica DMIRE2 microscope with a 63×/1.40–0.60 objective. Appropriate wavelengths for Cy3, Cy5, and Alexa-488 were used, and stacks of digital confocal images separated by 1 μm were acquired to visualize the pre- and postsynaptic components of the neuromuscular junction, as well as the terminal axons. Images were analyzed qualitatively with respect to the innervation pattern of acetylcholine receptors. PCA sections stained with hematoxylin–eosin were analyzed with conventional light microscopy (Leica).

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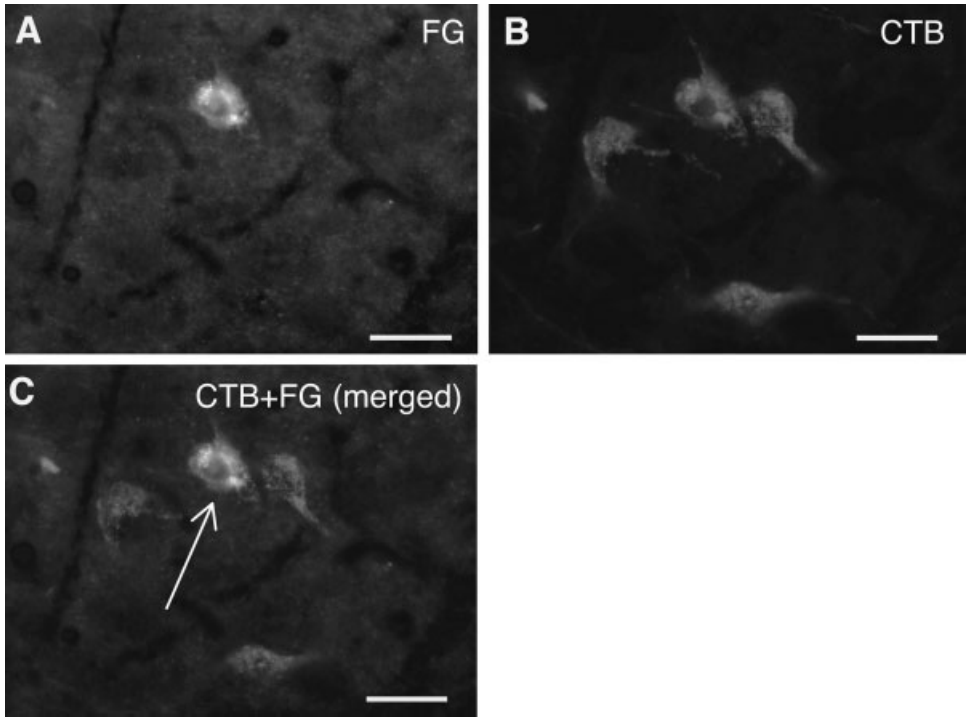


FIGURE 2. Micrographs from the same section of the brainstem at 6 weeks following chronic recurrent laryngeal nerve injury in the rostral nucleus ambiguus showing: **(A)** one neuron retrogradely traced from the posterior cricoarytenoid muscle (PCA) containing fluorogold; **(B)** superior laryngeal neurons retrogradely traced with cholera toxin subunit B, visualized by Cy3-fluorescent labeling; and (**C)** merged images, demonstrating colocalization of the two tracers in one neuron innervating the PCA through the superior laryngeal nerve (arrow). Bar = 30 μm.

RESULTS

Retrograde Tracing. FG-positive neurons projected to the PCA muscle, whereas CTB-positive neurons projected through the SLN. Hence: FG^+/CTB^- = neuron projecting through the RLN to the PCA; and FG^-/CTB^+ = neuron projecting through the SLN to other target than the PCA FG^+/CTB^+ = neuron projecting through the SLN to the PCA.

In normal controls, FG-positive neurons retrogradely traced from the PCA muscle were found in the NA (Fig. 2A), separated rostrocaudally into two groups (Fig. 3A). CTB-positive neurons (Fig. 2B) were found in the rostral NA, overlapping the area of the rostral group of FG-positive PCA neurons (Fig. 3A). Double-labeled cells (FG^+/CTB^+), representing SLN neurons projecting to the PCA muscle, were thus located only in the rostral part of the NA. The numbers of cells found in normal controls (mean ± SD) were as follows: FG^+/CTB^-, 27.2 ± 8.4; and FG^+/CTB^+, 11.6 ± 5.13. In addition, CTB-positive cells of smaller size were found in the dorsal motor nucleus of the vagus (not shown).

In animals subjected to chronic RLN injury, ipsilateral FG-positive neurons were found only in the rostral NA, with a preserved location compared with normal controls (Fig. 3B). As in the controls, the majority of these neurons were FG^+/CTB^+ (Fig. 2C). The numbers of cells found 6 weeks post-lesion (mean ± SD) were as follows: FG^+/CTB^-, 1.8 ± 3.6; and FG^+/CTB^+, 10.0 ± 3.74. The group of FG-positive PCA neurons in the caudal NA were thus depleted in chronically RLN-denervated animals (Fig. 3B), whereas the number of FG^+/CTB^+ neurons (i.e., ambiguus neurons projecting through the SLN to the PCA) remained intact (Fig. 3C). In the contralateral NA, occasional findings of FG-positive cells occurred in both normal control animals and chronically RLN-denervated animals.

Electrophysiology. In normal controls, the PCA showed an EMG pattern characterized by clear, phasic, inspiratory bursts (Fig. 4A). Stimulation of the RLN resulted in reproducible CMAPs of high amplitude (5.8 ± 1.0 mV). Stimulation of the SLN resulted in reproducible CMAPs of lower amplitude (1.4 ± 0.6 mV; Fig. 4B), thus representing a native innervation of the PCA by both the RLN and the SLN.

In the animals subjected to chronic RLN injury, residual inspiratory EMG activity was present (Fig.

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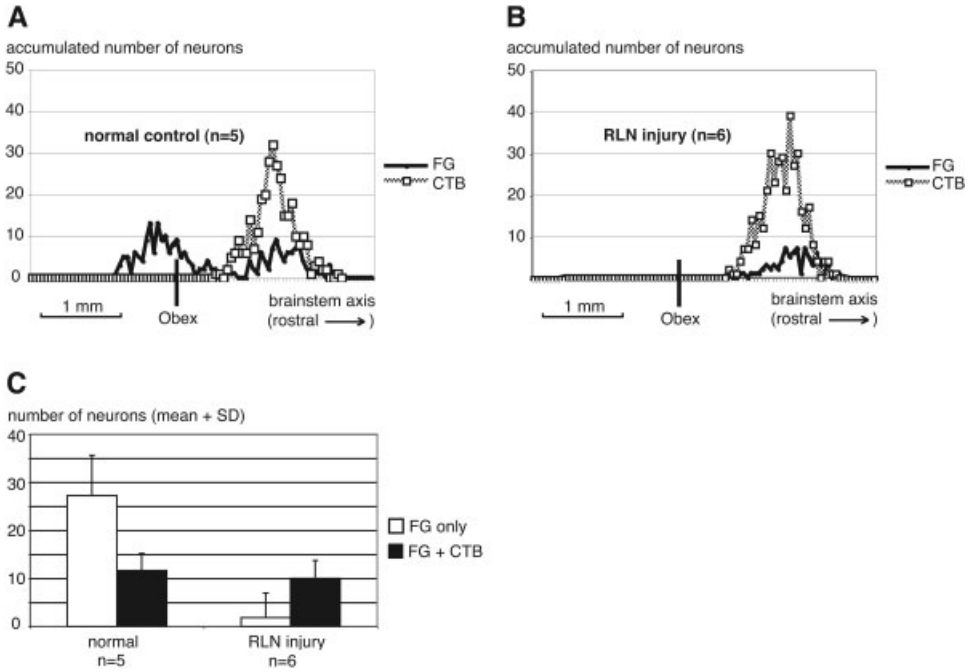


FIGURE 3. Rostrocaudal plot of the accumulated number of retrogradely traced neurons in the nucleus ambiguus following injection of fluorogold (FG) in the posterior cricoarytenoid muscle (PCA) (solid line), and application of cholera toxin B (CTB) to the superior laryngeal nerve (SLN) (dotted line). In normal controls **(A)**, the PCA neurons were found located in two distinct groups. In animals subjected to chronic recurrent laryngeal nerve (RLN) injury **(B)**, the rostral group of PCA neurons persisted. In both groups, the SLN neurons were found in the rostral nucleus ambiguus. **(C)** Bar plot demonstrating the number of neurons containing only FG and neurons colabeled with FG and CTB. No differences could be detected in the number of colabeled neurons between animals subjected to RLN injury and normal controls. This indicates an intact neuron pool that projects through the SLN to the PCA muscle. Neurons containing only FG were nearly depleted in injured animals.

4A). The amplitudes of the motor unit action potentials appeared lower in chronically denervated animals than in controls. No signs of denervation activity (i.e., fibrillation potentials and positive sharp waves) were detected during the electrophysiologically silent expiratory phase (Fig. 4A). Ipsilateral RLN stimulation did not result in any detectable CMAPs. Stimulation of the SLN resulted in reproducible CMAPs (2.2 ± 0.9 mV; Fig. 4B). The amplitude of the CMAPs following SLN stimulation was significantly higher in chronically RLN-denervated animals than in normal controls (P = 0,0242, unpaired t-test; Fig. 4C).

Muscle Analysis. Routine hematoxylin–eosin staining showed some muscle atrophy in the left PCA on the injured side (ipsilateral to the nerve injury) (Fig. 5A). On the uninjured side (contralateral to the nerve injury), morphologically clear synaptophysin (SP)-positive preterminal neuromuscular junctions were present (Fig. 5B). Ipsilateral to the nerve injury, SP immunoreactivity was dispersed, showing an increase in thin axonal-like structures (Fig. 5C), indicative of intramuscular sprouting. Due to the previous chronic RLN lesion, the origin of these fibers could not have been the ipsilateral RLN, but instead probably represented sprouting of fibers that reached the larynx through the SLN or the contralateral RLN.

On the uninjured side, confocal images of the PCA muscle demonstrated SP-positive and neurofilament (NF)-positive axons terminating on grape-like clusters of acetylcholine receptors (AChRs), as visualized with α-bungarotoxin (α-BT) (Fig. 6A). On the injured side (Fig. 6B), numerous thin SP-positive fibers with bouton-like morphology were seen running between the muscle fibers and terminating on the AChRs. There were also occasional NF-positive fibers innervating the AChRs.

DISCUSSION

In this study we have described the reinnervation of one of the intrinsic laryngeal muscles, the posterior cricoarytenoid muscle, following chronic left-sided injury to the recurrent laryngeal nerve. The results demonstrate electrophysiological evidence of dual innervation of the normal PCA, by the RLN and the SLN, with an amplitude ratio of about 4:1. We have also shown

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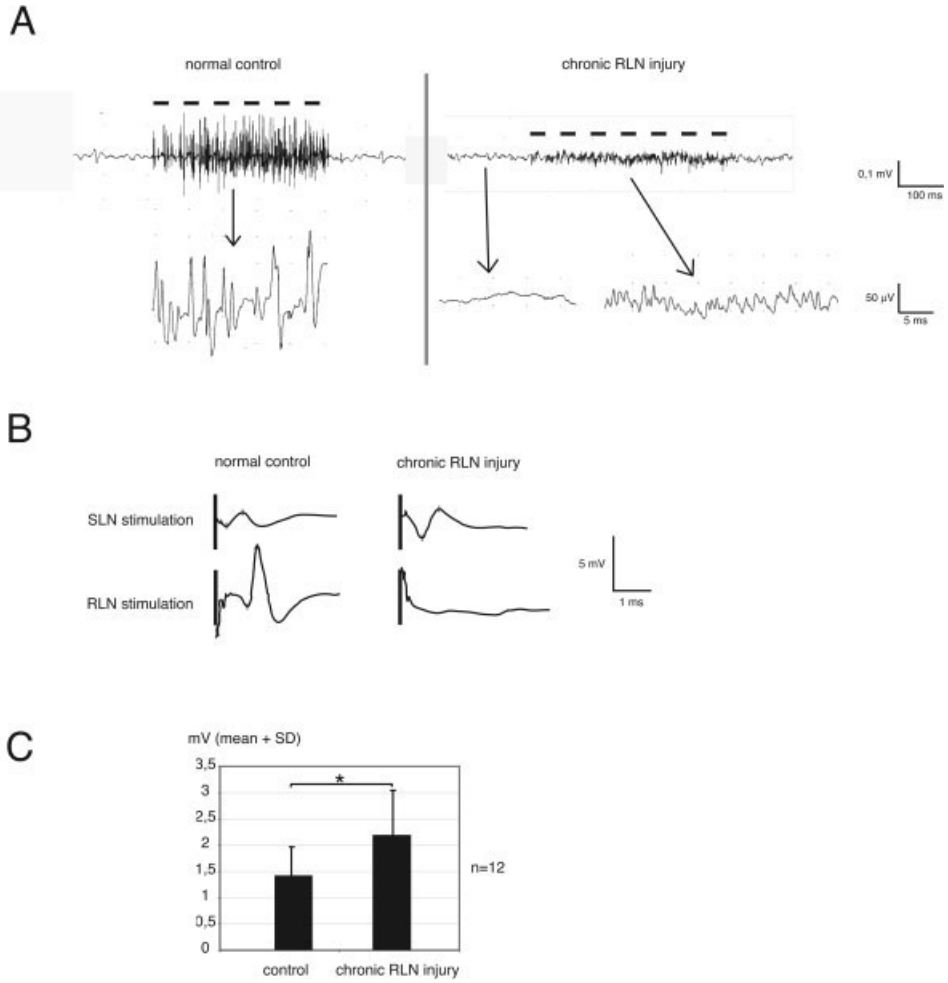


FIGURE 4. (A) Representative electromyogram (EMG) during spontaneous inspiratory activity of the posterior cricoarytenoid muscle (PCA) in normal controls and in animals subjected to chronic recurrent laryngeal nerve (RLN) injury. The inspiratory phase is marked by the dotted lines. **(B)** Compound muscle action potentials (CMAPs) recorded from the PCA following electrical stimulation of the superior laryngeal nerve (SLN) and RLN in normal controls and in chronically RLN denervated animals (6 weeks post-lesion). **(C)** Amplitude of CMAPs recorded in the left PCA following electrical stimulation of the left SLN in normal controls and in chronically RLN denervated animals, demonstrating an increase of SLN neuromuscular impact ($P = 0.0242$, *unpaired* t-*test).*

that, following RLN injury, the intact fibers of the SLN reinnervate the denervated PCA, most likely by intramuscular sprouting. This is the first study to show the functional importance of efferent collateral sprouting to an intrinsic laryngeal muscle.

The PCA is a paired muscle that rotates the arytenoid cartilages and widens the glottic aperture during inspiration. Generally it has been considered to be innervated exclusively by the RLN.[20,40] The location of the PCA motor neurons in rostral and caudal divisions of the nucleus ambiguus has been described previously by many investigators.[2,12,16,17,20,21,27–29] Portillo and Pásaro[29] demonstrated in the rat that the location of the rostral group of PCA neurons coincides with neurons projecting to the cricothyroid muscle, which is innervated by the SLN.[20] Thus, in this rostral part of the NA, neurons that innervate adductor and abductor muscles through both the RLN and the SLN are located in the same area. In the present study, double-labeled cells were found only in the rostral NA, in similar numbers and with similar locations in both normal and chronically RLN-denervated animals (Fig. 3). This shows that the PCA muscle receives innervation from one caudal group of neurons projecting through the RLN and one rostral group projecting through the SLN. Theoretically, this could be the result of uptake of fluorogold into the SLN, because the injection site in the PCA is in close proximity to Galen's anastomosis.[34] This is an unlikely scenario, however, because no leakage of tracer was observed, and the amount of leakage

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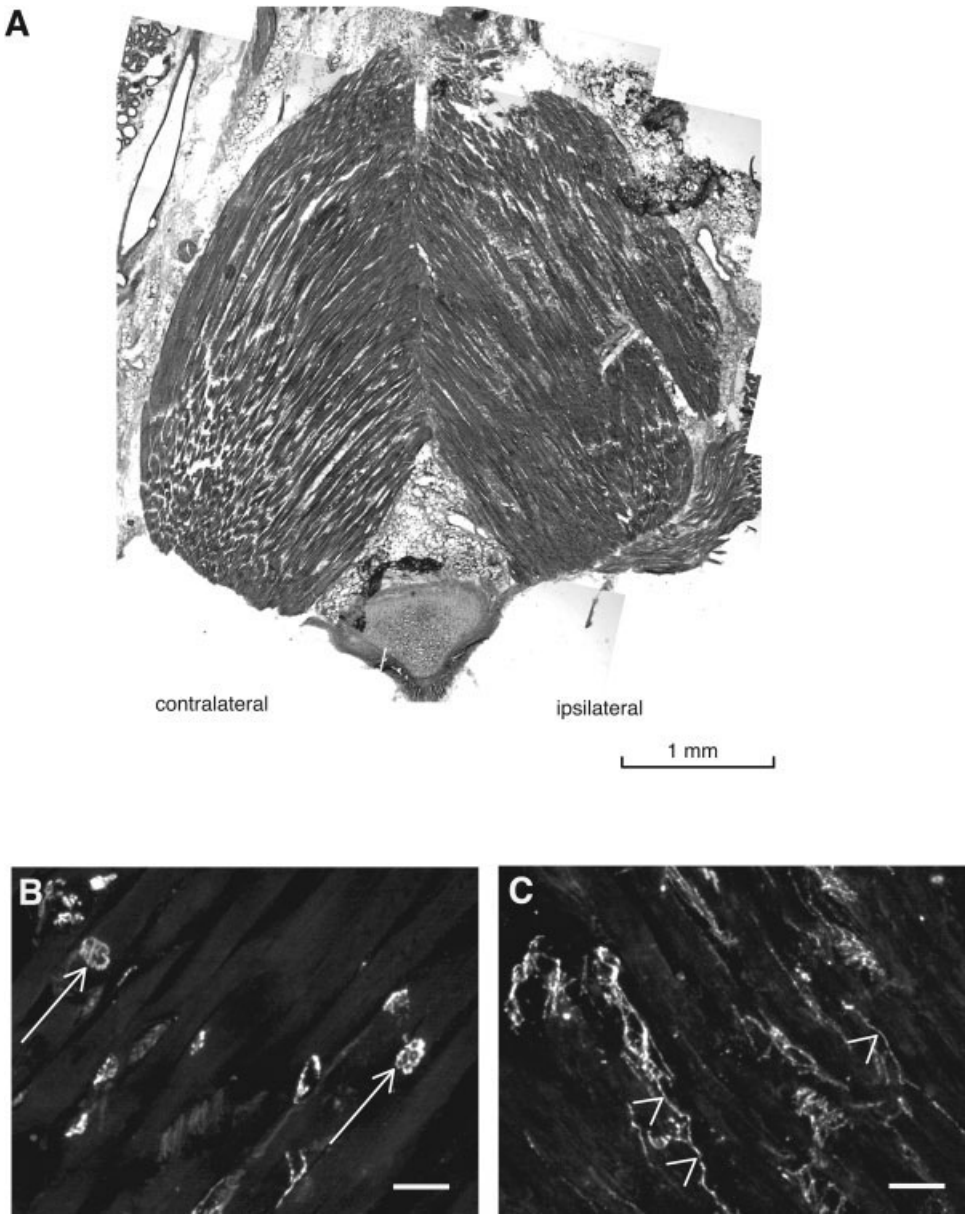


FIGURE 5. (A) Hematoxylin–eosin staining of a coronal section of the posterior cricoarytenoid muscle (PCA) at 6 weeks following chronic recurrent laryngeal nerve (RLN) injury. Ipsilateral to nerve injury, muscle fibers appear slightly atrophic. **(B, C)** Photomicrographs of the PCA 6 weeks after chronic RLN injury, immunolabeled for synaptophysin (SP). On the uninjured side **(B)**, morphologically clear SP-positive neuromuscular junctions are present (arrows). On the injured side **(C)**, SP immunoreactivity is more dispersed, labeling thin axonal structures running in between muscle fibers, probably representing intramuscular sprouting (arrowheads). Bar = 50 μm.

would have had to have been consistent in each animal. Furthermore, the electrophysiological finding that SLN stimulation resulted in a reproducible CMAP recorded from the PCA (Fig. 4B) supports the view that colocalized neurons represent SLN projections to the PCA. The RLN must be regarded as the main motor input to this muscle, as the majority of the PCA neurons are located in the caudal group in the NA in the RLN area.

Following stimulation of the SLN, the CMAP was significantly higher in animals subjected to chronic RLN injury than in normal controls (Fig. 4C). At the same time, there was no evidence of an increased number of double-labeled SLN PCA neurons (Fig. 3C). This indicates that no new neurons had been recruited, but intact pre-existing SLN fibers in the PCA had expanded their territories in response to RLN denervation, thus enlarging the motor units of

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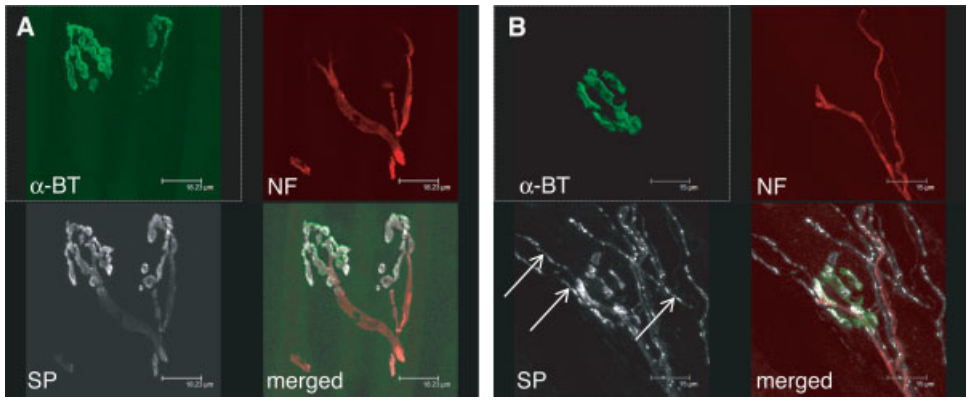


FIGURE 6. Confocal images of the posterior cricoarytenoid muscle at 6 weeks after chronic recurrent laryngeal nerve injury. On the uninjured side **(A)**, axons that innervate the grape-like clusters of acetylcholine receptors (AChRs) visualized with α-bungarotoxin (α-BT) are positive for both synaptophysin (SP) and neurofilament (NF). On the injured side **(B)**, thin SP-positive fibers with bouton-like morphology (arrows) run in between the muscle fibers to the AChRs and terminate on them. In addition, one terminal axon positive for both SP and NF is seen innervating the cluster of AChRs.

the SLN. The change in CMAP amplitude (from 1.4 to 2.2 mV) could be regarded as a rather moderate degree of collateral reinnervation. Data obtained from partial denervation studies demonstrate that motor neurons increased their innervation fields several-fold up to 1.5 years post-lesion.[14,22,30] However, the present study used a shorter post-operative survival time than was the case in these experiments, and a later increase in CMAP amplitude cannot be ruled out. The CMAP amplitude also depends on the state of the PCA muscle, which in this model showed signs of slight atrophy (Fig. 5A) resulting from the total deprivation of ipsilateral RLN input. Furthermore, the SLN fibers probably compete with other neural input for reinnervating the neuromuscular junctions in the PCA, including sympathetic nerve fibers, an effect that has been demonstrated previously.[25,26]

Following injury to the RLN, the phenotypic characteristics of the intrinsic laryngeal muscles undergo specific changes,[35] which also correlate with the degree of vocal fold motion impairment.[36] These changes have been proposed as being secondary to neural regulation; that is, that reinnervation by a motor neuron with different firing properties than the original leads to a change in the expression of specific myosin heavy chains in the muscle cell, which then affects muscle speed and power output.[31] This has also been used as an experimental tool for judging the degree of accurate reinnervation after RLN injury.[19] The results obtained from these studies show that, following RLN transection, the fibers of the rat TA and PCA muscles eventually shift to a slower phenotype, with elevated expression of myosin heavy chain type 2A and 2X, but lower levels of type 2B.[31,36] In other words, there is a shift toward a phenotype similar to the slower cricothyroid muscle, which is phenotypically different and is innervated by the SLN. These results would therefore support the current findings of SLN collateral reinnervation after RLN transection.

Upon denervation of a skeletal muscle, acetylcholine receptors upregulate and start spontaneous activity, resulting in characteristic EMG findings (positive sharp waves and fibrillation potentials), which are present from around 2 weeks post-injury until reinnervation has occurred. The present study revealed no such signs in the PCA muscle at 6 weeks after chronic RLN injury. Instead, clear inspiratory EMG activity was present, with motor unit action potential amplitudes apparently lower than those in normal control animals (Fig. 4A). In an additional control experiment in one normal rat, this residual EMG activity was abundant following acute ipsilateral nerve injury. It was nonexistent when both RLNs were transected, indicating that this EMG activity is due to contralateral RLN fibers crossing the midline. This is another possible source for reinnervation of PCA muscle fibers. The limited crossing of some RLN axons across the midline is further supported by the finding of occasional FG-positive neurons in the contralateral nucleus ambiguus. The current data suggest that the SLN fibers responsible for collateral reinnervation are silent during this spontaneous respiratory activity, which requires activation from higher brainstem centers, but visible as enlarged CMAPs following supramaximal electrical stimulation of the nerve. Sources of potential error include distant uptake by the hook-wire electrode from the contralateral PCA or the cricothyroid mus-

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cle; however, this is not likely to have occurred, because the uptake of electrical activity is confined to the area just around the electrode. Furthermore, the location of the electrode within the left PCA was confirmed macroscopically during surgery and also microscopically in muscle sections.

The presence of intramuscular sprouting was indicated by proliferation of SP-positive nerve fibers in the PCA muscle sections ipsilateral to the nerve injury (Fig. 5C). SP is a synaptic vesicle docking protein, present in the preterminal portion of the neuromuscular junction, but it is also present to some extent in the axons to which the protein is transported. Morphologically, these newly formed SP-positive fibers were thin and contained boutons. They ran between muscle fibers to make contact with the AChRs (Fig. 6B), and they were also sometimes mature enough to express neurofilament (Fig. 6B). Because the ipsilateral RLN was subjected to chronic injury and was directed away from the larynx, the most probable origin of this innervation is the SLN, a view supported by the electrophysiological findings, although the contralateral RLN may also have been involved. A detailed analysis of the specific change of projections to the PCA neuromuscular junctions would require anterograde tracing of the laryngeal nerves, but this was not performed in the present study.

Following RLN axonal injury, there is a retrograde reaction that includes synaptic detachment from brainstem neurons and a loss of cortical contact.[4] Cortical contact with the brainstem neurons is re-established if they succeed in reinnervating the target, and the result may be a different cortical representation. Although it has been suggested that there is a low degree of misguided reinnervation following crush injury,[11] the somatotopic organization in the cortex will be different. This will in turn necessitate central plasticity. The results of the present study demonstrate that collateral reinnervation is part of the pathology following RLN injury, which also results in altered cortical representation of the PCA. On this basis, we speculate that insufficient cortical adjustment after RLN regeneration as well as SLN collateral reinnervation following RLN injury may be an important factor for the uncoordinated vocal cord movements seen after these lesions.

In conclusion, this study has shown, by retrograde tracing and electrophysiological methods, that fibers from the SLN functionally reinnervate the PCA muscle after RLN injury. This may be important when choosing surgical and/or medical treatment for patients with RLN injury.

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